DIPLOMA PROGRAMME PRACTICE · SL · SET A

Biology / Paper 2

Worked mark scheme · 50 marks · 90 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each. Answer all of Section A and one of two Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 1 [12]

Shell variation across a town

(a) Calculate the percentage of yellow snails in each habitat and express the difference in percentage points. [3]

1 Urban: 150/240×100=62.5%.

1 Woodland: 96/240×100=40.0%.

1 Difference is 22.5 percentage points (urban higher).

(b) Distinguish a colour category from a banding category using the photographs. Why would recording colour alone lose information? [2]

1 Pink and yellow describe background colour whereas banding describes presence or number of dark bands.

1 The two pink shells shown have different banding patterns, so pooling them hides variation.

(c) Propose a selection mechanism that could produce the urban pattern. State a necessary condition for a response across generations. [3]

1 Under equivalent exposure, a paler shell could absorb less radiation, reducing heat stress; accept a supported alternative involving predators.

1 If surviving individuals leave more offspring, the colour-associated variants may increase in frequency.

1 Differences in shell colour must have a heritable component.

(d) Why would treating all 240 snails in one habitat as independent temperature replicates exaggerate the evidence? Propose a site-level analysis. [2]

1 Snails within one site experience shared conditions and are nested within that site.

1 Calculate each site's yellow proportion and compare it with that site's temperature across the independent sites.

(e) Design a comparison that tests a thermal effect of colour while controlling a plausible confounder. [2]

1 Expose comparable shells or live snails of different colours to standardized light and measure temperature change over the same interval.

1 Match or randomize banding, size, orientation or substrate; repeat for multiple independent specimens.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 2 [8]

Carbon uptake and cell respiration

(a) Calculate gross oxygen production in light. Explain the need for the dark measurement. [3]

1 Gross rate=2.4+0.6=3.0 mg O_2 h^{-1}.

1 In light, photosynthesis produces oxygen while respiration consumes some simultaneously.

1 The dark measurement estimates consumption when photosynthetic production is absent.

(b) Explain how the chloroplast and mitochondrion contribute to carbon cycling within these cells. [3]

1 Photosynthesis incorporates inorganic CO_2 into organic compounds using light energy.

1 Respiration breaks down organic substrates, releasing CO_2.

1 The carbon released can become available for photosynthesis again.

(c) Give a reason the dark correction could be biased and a control for an abiotic change in measured oxygen. [2]

1 Respiration in darkness may differ from respiration in light, or temperature may change during measurement.

1 Run the same illuminated and dark solutions without algae to estimate sensor drift or physical gas exchange.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 3 [7]

Transport from lung to artery

(a) Calculate cardiac output in dm^3 min^{-1} for both conditions. [2]

1 Rest=68×72/1000=4.896 dm^3 min^{-1}.

1 Exercise=95×132/1000=12.54 dm^3 min^{-1}.

(b) Use the detection times to reconstruct the tracer's route from the lung capillaries to the aorta. Identify where each of the two valves is crossed. [3]

1 Pulmonary veins carry the oxygenated blood to the left atrium.

1 It passes through the left atrioventricular (mitral) valve to the left ventricle.

1 It leaves through the aortic semilunar valve into the aorta.

(c) Explain why elevated cardiac output alone cannot establish that muscle oxygen consumption rose by the same factor. [2]

1 Consumption also depends on the difference in oxygen content between arterial and venous blood.

1 The fraction of total blood flow reaching active muscles can change.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 4 [7]

From coding sequence to phenotype

(a) Give the original mRNA sequence and amino acid sequence with direction of reading indicated. [2]

1 5′-GCU UUU GAA-3′, read 5′ to 3′.

1 Ala–Phe–Glu.

(b) Identify the mutation type at the nucleotide level and its effect on the polypeptide segment. [2]

1 It is a single-base substitution (T→C in the coding strand).

1 The change is synonymous since UUC and UUU both encode phenylalanine, so this segment's amino acid sequence is unchanged.

(c) Compare the likely consequence of deleting the first T of the original TTT triplet. Explain why DNA sequence alone cannot establish the size of a phenotypic effect. [3]

1 A one-base deletion shifts the grouping of downstream codons.

1 It can alter downstream amino acids or introduce an early stop.

1 Phenotypic outcome also depends on protein function, position, expression or compensation by another allele.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 5 [16]

Keeping pathogens out and remembering infection

(a) Outline three ways barriers reduce entry or establishment of pathogens. [3]

1 Mucus traps particles and microorganisms.

1 Cilia move trapped material toward removal from the airway.

1 Intact epithelium forms a physical barrier; accept a suitable additional mucosal or chemical defence.

(b) Explain how a phagocyte helps contain microbes that have crossed a barrier. [5]

1 It recognizes or attaches to a microbe.

1 It surrounds the microbe with its membrane.

1 A vesicle containing the microbe forms by engulfment.

1 It fuses with a lysosome or receives hydrolytic enzymes.

1 The microbe is digested or destroyed within the cell.

(c) Explain a specific antibody response to an antigen and why re-exposure can produce a faster response. [7]

1 A particular antigen is recognized by a lymphocyte with complementary specificity.

1 Activation selects specific clones, with helper T-cell signals where appropriate.

1 Selected B cells divide by mitosis, expanding the clone.

1 Some differentiate into plasma cells.

1 Plasma cells secrete antibodies specific to the antigen.

1 Binding can neutralize the pathogen or facilitate its removal.

1 Some cells remain as memory cells and rapidly generate a larger response on re-exposure.

Paper 2 · Question 5 (continued)

(Quality) Quality of the complete extended response [1]

1 Award one mark when the complete response communicates the relevant biological reasoning clearly and concisely, with logical connections and little repetition. Assess this across the parts, independently of the content score.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 6 [16]

A woodland after a storm

(a) Outline how three different feeding roles obtain organic matter in the woodland. [3]

1 Producers fix inorganic carbon to make organic compounds.

1 Consumers ingest other organisms or material derived from them.

1 Saprotrophs digest dead organic material externally and absorb soluble products.

(b) Explain why energy available to a secondary consumer is less than the chemical energy originally captured by producers. [5]

1 Some producer biomass is never ingested by primary consumers.

1 Some ingested material is not assimilated and leaves as faeces.

1 Respiration transfers usable energy from organic compounds and dissipates heat.

1 Respiratory and other losses also occur at the next trophic transfer.

1 Only the fraction retained as new biomass is available for a subsequent consumer to incorporate.

(c) Discuss how carbon moves between the atmosphere, fallen wood, decomposers and regrowing trees over time. [7]

1 Atmospheric CO_2 diffuses into leaves.

1 Photosynthesis incorporates its carbon into organic compounds.

1 Trees use some assimilated carbon to make structural biomass such as cellulose.

1 Fallen wood retains previously fixed carbon after the tree dies.

1 Decomposer enzymes break down wood and some carbon enters decomposer biomass.

1 Decomposer and tree respiration return carbon to the atmosphere as CO_2.

1 Net storage depends on the balance between fixation and release over the period, so regrowth alone does not prove a net sink.

(Quality) Quality of the complete extended response [1]

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